A PROJECT BASED SEMINAR REPORT

on

TicketFlowAI

Intelligent Auto-Handling of Support Tickets with Confidence-Based Human-in-the-Loop (HITL)

Submitted towards the
Partial Fulfilment of the Requirements of

Bachelor of Technology in Computer Engineering

By

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CERTIFICATE

This is to certify that the seminar report entitled

TicketFlowAI

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being submitted by

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is a bonafide work carried out by student under the supervision of Prof. J. R. Mankar and it is submitted towards the partial fulfilment of the requirement of Bachelor of Technology (Computer Engineering) during academic year 2025-2026.

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Abstract

In the helpdesk systems, modern organizations get customer and IT support requests in large numbers in form of service tickets. The management of these tickets is also worthwhile in order to get the issues resolved promptly and to sustain the quality of services. Conventional ticket management systems are dependent on human engineers who will review, categorize, prioritize and assign tickets manually. With more support requests, this manual system is time consuming, the support requests are not addressed in time, which results in higher workload, decreased operational efficiency and efficiency. The offered system presents a smart system of handling support tickets that analyses the description of the ticket automatically, recognizes the types of issues and their priority and finds the solution to these issues by consulting solved cases. Semantic similarity search is another feature that is performed to find historical data related to the new ticket in the system. The easy and routine problems may be resolved automatically and the complicated cases are passed on to the human specialists by one of the human-assisted decision processes. Such solution is beneficial as it minimizes manual workload, increases the speed of resolving the tickets, and increases support services efficiency and reliability in any industry.

Keywords: *Support Ticket Management, Service Desk Automation, Semantic Similarity Search, Knowledge Base Retrieval, Human-Assisted Decision Systems, Intelligent Service Operations.*

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Mst. Swayam Vijay Dusing

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Chapter 1

Introduction

1.1 Project Idea

The proposed project, TicketFlow, is an intelligent IT Service Management (ITSM) system designed to automate the classification, prioritization, and routing of support tickets using Natural Language Processing (NLP) and machine learning. Unlike traditional systems that rely on basic sorting, this platform utilizes a multi-task decision engine to simultaneously evaluate ticket categories, urgency levels, and potential Service Level Agreement (SLA) risks. By integrating semantic vector search via FAISS and transformer models like BERT, the system can automatically resolve repetitive issues with high precision while providing transparent, narrative explanations for its decisions through SHAP-based explainability.

To ensure reliability in complex enterprise environments, the system employs a Human-in-the-Loop (HITL) mechanism based on a calculated confidence score. When the system's confidence falls below a specific threshold, or when a ticket is identified as high-risk or novel, it is automatically escalated to a human expert for review. This hybrid approach reduces the manual workload for support teams and minimizes resolution times, while the continuous feedback loop allows the model to learn from human corrections, ultimately improving service reliability and customer satisfaction.

1.2 Motivation

The increasing volume of service requests in modern organizations often creates significant bottlenecks in manual ticket handling, leading to delayed resolution times and increased operational costs. As IT departments and helpdesks face growing pressure to maintain service quality, there is a critical need for intelligent systems that can automate routine tasks and prioritize urgent issues with high precision. Traditional manual sorting is not only time-consuming but also prone to human error, which can negatively impact overall service reliability and user satisfaction.

By leveraging advanced computational techniques, organizations can transition toward a more proactive support model that ensures resources are directed where they are most needed. The integration of automated decision-making—balanced by human oversight—allows for a scalable solution that maintains high standards of accuracy and transparency. Ultimately, the drive for such a system stems from the objective to enhance institutional efficiency, reduce the workload on technical personnel, and ensure that critical service level commitments are consistently met.

1.3 Literature Survey

Over the past few years, significant progress has been made in the Automated processing of IT service requests. However, many existing systems remain either domain-specific or lack the integrated intelligent mechanisms necessary for handling the high volume of unstructured data while maintaining transparency and strict service level adherence. This literature review explores several notable contributions in the field and highlights the research gaps that this project aims to address.

Accurate classification and secure management of service level agreements (SLAs) are critical components in modern helpdesk environments. Traditional techniques often struggle with the complexity of unstructured text and the need for audit-ready compliance. Krzysztof Bocianiak et al. proposed a security-focused approach to SLA management specifically for cloud systems, identifying security monitoring and compliance as key pillars. While their theoretical foundation for "Proof of Security" is robust, the model focuses primarily on conceptual frameworks rather than practical, measurable ticket classification. Our system operationalizes these concepts by integrating SLA risk prediction directly into the multi-task decision engine.

To improve user engagement and reduce manual sorting loads, research by Esada Licina et al. investigated the optimization of ticketing systems through community platform integration, such as Discord. Their findings indicated high user satisfaction and improved real-time communication. However, the approach is highly dependent on active community participation and was tested on a limited sample size. This project extends the goal of reducing manual effort by implementing a robust backend using FastAPI and MongoDB to handle high-volume enterprise requests independently of community-driven platforms.

Further advancements in transparency were explored by Esada Licina et al. through the development of "HelpAgent," an agent-based system utilizing Large Language Models (LLMs) and SHAP for explainable AI. While their research successfully demonstrated that narrative explanations improve user trust, the system remained in an early poster-work stage with a complex interface. This project bridges that gap by implementing Mistral-Nemo via Ollama for efficient local LLM processing and using SHAP to provide clear, automated justifications for classification decisions within a minimalist React-based interface.

Finally, the technical efficiency of multi-category classification was addressed by M. Ishwarya Niranjana et al. , who utilized the XGBoost algorithm to achieve high accuracy in handling unstructured support text. Despite the strong performance metrics, such models can be sensitive to dataset imbalances and lack a human-centric fallback. Our system mitigates these limitations by incorporating a Human-in-the-Loop (HITL) mechanism that uses a weighted confidence score—aggregating category, priority, and SLA scores—to ensure that low-confidence or high-risk cases are escalated to experts for final approval.

In summary, while the existing literature offers strong foundations in NLP classification, security-focused SLAs, and explainable AI, this project advances these technologies by tightly integrating automated multi-task decision-making, semantic vector search via FAISS, and a confidence-based escalation framework into a unified, scalable ITSM solution.

Chapter 2

Problem Definition and scope

2.1 Problem Statement

2.1.1 Aim

To design and develop a system that automatically analyzes, classifies, prioritizes, and routes customer and IT support tickets using Natural Language Processing (NLP) and machine learning so that support operations become faster, more accurate, and less dependent on manual effort, while ensuring reliable outcomes through a Human-in-the-Loop (HITL) escalation mechanism.

2.1.2 Goals and Objectives

The primary goal of this project is to design and develop an intelligent, risk-aware IT Service Management (ITSM) system that automates the lifecycle of support tickets to enhance operational efficiency. The specific objectives include:

1. **Study and Identification:** To evaluate existing ticket management systems and identify current limitations regarding automation, response efficiency, and intelligent resolution.
2. **Architectural Design:** To design a comprehensive system architecture that integrates data processing, machine learning models, and automated decision mechanisms.
3. **System Development:** To develop core components, including a user-friendly interface for ticket submission, tracking, and a backend capable of both automated and human-assisted resolution.
4. **Validation and Testing:** To test the system in a simulated environment, evaluating the accuracy of ticket classification, priority detection, and the effectiveness of Human-in-the-Loop (HITL) handling.
5. **Prototype Deployment:** To deploy a functional prototype that demonstrates real-world automated ticket handling and seamless escalation workflows.

2.1.3 Assumptions and Scope

Assumptions

- **Data Availability:** It is assumed that historical ticket data is available and sufficiently labeled to train the initial machine learning and transformer models.
- **User Competency:** It is assumed that end-users will provide descriptive ticket titles and summaries to allow for effective NLP-based analysis.
- **Infrastructure Reliability:** The system assumes constant connectivity to the vector database (FAISS) and local LLM services for real-time processing and similarity searches.

Scope

- **Ticket Types:** The system is designed to handle common IT service requests, including authentication issues, network disruptions, hardware/software errors, and security-related access requests.
- **Automation Boundaries:** The project focuses on "Intelligent Auto-Handling," where repetitive known issues are resolved automatically only when confidence scores meet a predefined high threshold.
- **Human-in-the-Loop:** The scope includes a dedicated escalation mechanism for low-confidence, high-risk, or time-sensitive tickets.
- **Target Environment:** The system is primarily tailored for IT departments, corporate helpdesks, and SaaS support teams.

Chapter 3

Methodology

3.1 Methodology

Refer to Fig. 3.1 which shows the complete block diagram explaining the overall process flow of the system.

1. **Ticket Submission:** Users submit support tickets through the system interface, providing details such as title, description, and user-specific information.
2. **Preprocessing and Text Analysis:** The submitted ticket undergoes preprocessing including text cleaning, tokenization, and normalization. Natural Language Processing (NLP) techniques are applied to prepare the data for further analysis.
3. **Multi-Model Classification:** Machine Learning and Transformer-based models (such as Sentence Transformers and BERT) are used to classify the ticket into appropriate categories, predict priority levels, and estimate SLA risk.
4. **Semantic Similarity Search:** The processed ticket is converted into vector embeddings and compared with historical ticket data using FAISS (Facebook AI Similarity Search) to retrieve similar past tickets and their resolutions.
5. **Confidence Score Calculation:** A confidence score is computed using a weighted combination of multiple model outputs, including category prediction, priority estimation, and SLA risk probability. This score determines the reliability of automated decisions.
6. **Decision Engine (Automation vs HITL):** Based on the confidence score:
   - High confidence: The system automatically resolves the ticket using retrieved solutions.
   - Medium confidence: Suggestions are provided to the support agent for review.
   - Low confidence: The ticket is escalated to a human expert.
7. **Human-in-the-Loop (HITL) Processing:** For low-confidence or high-risk tickets, human agents review, modify, or approve the suggested solution to ensure accuracy and reliability.

8. **Response Generation and Delivery:** The final resolution, either automated or human-approved, is sent back to the user through the system.
9. **Storage and Feedback Loop:** All tickets, decisions, and resolutions are stored in the database. Human corrections are captured and fed back into the system to improve model performance over time.
10. **Model Retraining and Continuous Improvement:** The system periodically retrains its models using updated data to enhance classification accuracy, confidence estimation, and overall system efficiency.

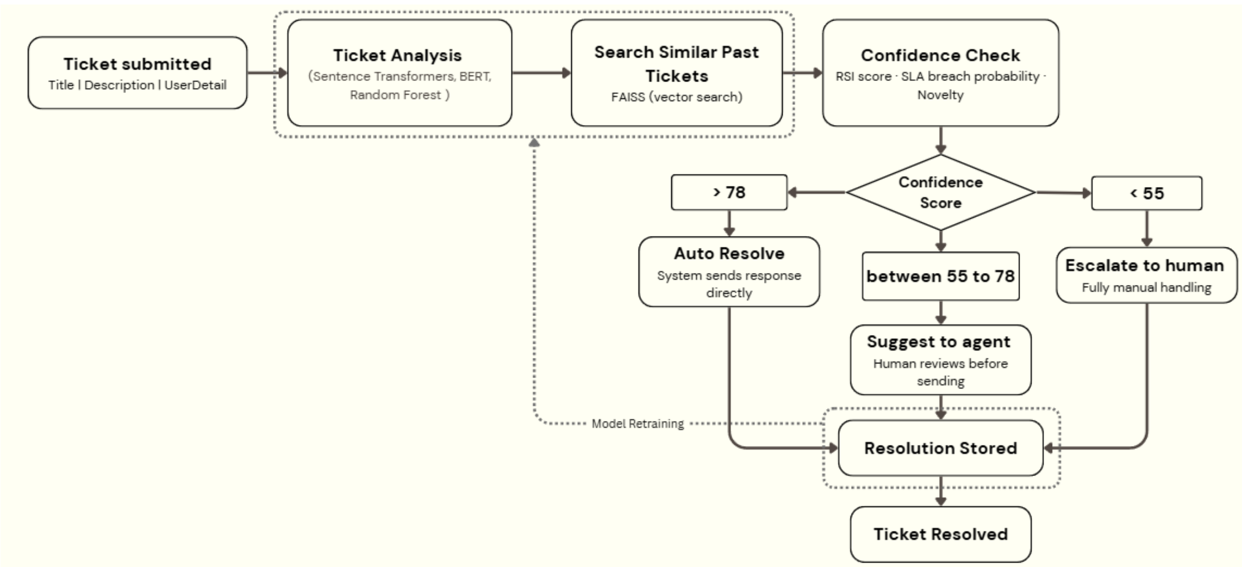

Figure 3.1: Block Diagram of TicketFlowAI System

Chapter 4

Outcome

4.1 Type of Project

The TicketFlowAI project is primarily application-oriented, with a focus on practical implementation and real-world problem-solving within the domain of intelligent IT service management and support ticket automation.

- **Application-Oriented:**
  The core goal of TicketFlowAI is to develop a functional and deployable system that can be used by organizations such as IT departments, helpdesks, and SaaS support teams to automate the handling of support tickets. The system focuses on reducing manual effort by intelligently analyzing, classifying, prioritizing, and resolving tickets. By leveraging Natural Language Processing (NLP), Machine Learning, and Human-in-the-Loop (HITL) mechanisms, the project aims to improve efficiency, reduce response time, and ensure reliable decision-making. The system is designed for practical use with a user-friendly interface and seamless integration into existing workflows.

- **Domain-Specific:**
  - **IT Service Management:** TicketFlowAI is specifically designed for IT service environments, enabling efficient handling of issues such as authentication failures, network disruptions, software errors, and access requests.
  - **Artificial Intelligence:** The system utilizes NLP, transformer-based models, and machine learning algorithms to perform ticket classification, priority detection, and SLA risk prediction.
  - **Data Engineering & Retrieval:** The project incorporates vector-based similarity search (FAISS) to retrieve relevant past tickets and resolutions, enabling intelligent and context-aware automation.

- **Research-Oriented:**

While the primary focus of TicketFlowAI is on application, the project also incorporates research elements in improving intelligent decision-making in ticket automation systems. It explores confidence-based automation using weighted model outputs, combining classification, prioritization, and SLA prediction into a unified decision engine. Additionally, the integration of Human-in-the-Loop (HITL) introduces a hybrid approach that balances automation with human expertise. The project also investigates the effectiveness of semantic similarity search and feedback-driven model improvement, contributing to advancements in AI-driven service management systems.

Chapter 5

Project Plan

5.1 Project Timeline

your project timeline from jan 2025 to Dec 2025

5.2 Team Organization

5.2.1 Team structure

Each Team Members Contribution/work distribution in Detail

Chapter 6

Experimental setup

In order to rigorously evaluate TicketFlowAI, we established a controlled experimental environment that simulates real-world IT service management conditions where support tickets are generated and processed. Below are the details of our setup, including the dataset, technologies used, and performance parameters considered during evaluation.

6.1 Data Set

Custom Ticket Dataset

- **Source:** A collection of 1000+ support tickets obtained from publicly available IT helpdesk datasets and synthetically generated data to simulate real-world scenarios.
- **Formats:** Structured and semi-structured text data including ticket titles, descriptions, user metadata, and category labels.
- **Content Variety:**
  - Authentication issues (e.g., login failure, password reset)
  - Network-related problems (e.g., VPN issues, connectivity loss)
  - Hardware and software errors
  - Access and permission requests
  - General IT and service-related queries

6.2 Technology Used

- **Frontend:**
  - React 18.2 with Tailwind CSS for building a responsive and user-friendly interface
- **Backend & Infrastructure:**
  - FastAPI (Python) for handling API requests and system logic
  - MongoDB (NoSQL) for storing ticket data, results, and logs
- **Natural Language Processing & ML Models:**

  - Sentence Transformers for semantic embedding generation
  - BERT / Machine Learning models (e.g., Random Forest) for classification and prediction tasks

- **Vector Search:**
  - FAISS (Facebook AI Similarity Search) for retrieving similar past tickets based on embeddings
- **LLM & Explainability:**
  - Mistral-Nemo (via Ollama) for generating responses and explanations
  - SHAP for model explainability and interpretation
- **Development Tools:**
  - Git for version control
  - Docker for environment consistency (optional deployment setup)

6.3 Performance Parameters

To evaluate system performance and effectiveness, the following parameters were considered:

- **Classification Accuracy:**
  - Accuracy of ticket categorization using ML models
  - Target: $\geq 85\%$
- **Priority Prediction Accuracy:**
  - Accuracy in identifying correct priority levels (Low, Medium, High, Critical)
  - Evaluated using Precision, Recall, and F1-score
- **SLA Risk Prediction:**
  - Ability to correctly predict tickets likely to breach SLA deadlines
  - Measured using classification metrics (Precision, Recall, F1-score)
- **Confidence Score Effectiveness:**
  - Accuracy of decision-making based on confidence thresholds (Auto vs HITL)
  - Evaluated by comparing automated decisions with human-reviewed outcomes
- **Response Time (Latency):**
  - Time taken from ticket submission to final resolution output
  - Target: ≤ 5 seconds per ticket
- **System Throughput:**
  - Number of tickets processed per minute
  - Used to evaluate scalability of the system
- **Human-in-the-Loop Efficiency:**
  - Percentage of tickets correctly escalated to human agents

  - Reduction in manual workload compared to fully manual systems

- **Similarity Retrieval Accuracy:**
  - Effectiveness of FAISS in retrieving relevant past tickets
  - Evaluated using relevance metrics (Top-K accuracy)

Chapter 7

PO & PSO Mapping

7.1 Program Outcome (PO) Mapping

PO	Description	Level	Justification
PO1	**Engineering knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals to AI and DS problems.	3	ArchieveMind applies engineering and AI concepts (OCR, NLP, blockchain) to solve complex legacy data extraction problems.
PO2	**Problem analysis:** Identify, formulate, and analyze complex problems using principles of math, natural science, and engineering.	3	Legacy data inconsistency and extraction issues were analyzed deeply using AI/NLP frameworks.
PO3	**Design/development of solutions:** Design system components or processes considering societal, safety, and environmental factors.	2	Created a rough design using AI, but actual coding hasn't started yet.
PO4	**Conduct investigations of complex problems:** Conduct research-based investigations, design experiments, analyze data, and synthesize conclusions.	3	Literature review and OCR/NLP model experiments were done; analysis was data-driven.

PO5	**Modern tool usage:** Create and apply modern tools, modeling, and engineering techniques.	3	Modern tools like OCR, AI models, decentralized storage, blockchain, Puppeteer API are extensively applied in ArchieveMind.
PO6	**The engineer and society:** Apply contextual knowledge to societal, health, safety, legal, and cultural issues.	2	Focused on data security and compliance (sensitive sectors: healthcare, banking) but less direct involvement with law or culture.
PO7	**Environment and sustainability:** Understand societal/environmental impact and sustainable development.	1	Indirect impact: project digitizes records reducing physical storage/paper but not a major environmental project.
PO8	**Ethics:** Apply ethical principles and commit to ethics and professional responsibilities.	3	Blockchain logging, data privacy, secure access control directly reflect ethical handling of data and transparency.
PO9	**Individual and team work:** Function effectively as an individual and in teams.	3	Developed under mentorship and collaboration demonstrating both individual and team efforts.
PO10	**Communication:** Communicate effectively on complex engineering activities.	2	Maintained clear project documentation, presentation (PPT), and technical reporting in the ArchieveMind project.
PO11	**Project management and finance:** Apply project management and financial principles in engineering work.	2	The project involved preliminary planning for aspects such as decentralized storage costs, secure cloud uploads, and blockchain integration; however, it did not encompass comprehensive financial management activities typical of a full-scale engineering project.

PO12	**Life-long learning:** Recognize the need for life-long learning amidst technological change.	3	Continuously learning and adapting AI/ML/OCR/NLP advancements for better project outcomes.

Table 7.1: Program Outcomes (PO) Mapping – ArchieveMind

7.2 Program Specific Outcome (PSO) Mapping

PSO Number	**Description**	**Mapping Level**	**Justification**
PSO1	Interpret data and apply AI algorithms to provide solutions.	3	Applied AI algorithms (OCR, NLP, fuzzy matching, blockchain security) to solve legacy data extraction and structuring.
PSO2	Apply standard practices, strategies, and models of Data Science to discover knowledge.	3	Applied standard practices (AI pipeline, OCR-NLP processing, deduplication, secure storage) to transform unstructured data into structured knowledge.

Table 7.2: Program Specific Outcomes (PSO) Mapping – ArchieveMind

Chapter 8

Summary

TicketFlowAI offers a robust AI-powered solution for automating the handling of support tickets in IT service management systems. By integrating Natural Language Processing (NLP), Machine Learning, semantic similarity search, and a confidence-based Human-in-the-Loop (HITL) mechanism, the system ensures both efficiency and reliability in ticket resolution. It intelligently analyzes incoming tickets, classifies them, predicts priority and SLA risks, and retrieves relevant solutions using vector-based search (FAISS).

The system performs automated resolution for high-confidence cases while escalating complex or uncertain tickets to human experts, ensuring accuracy and reducing operational risks. With the use of transformer-based models and a unified decision engine, TicketFlowAI enables faster response times, reduced manual workload, and improved service quality. Additionally, the feedback loop mechanism allows continuous learning and model improvement over time.

Overall, TicketFlowAI provides a scalable, intelligent, and reliable platform for modern support ticket automation, enhancing both system performance and user satisfaction.

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